

Leonard C. Lupin-Jimenez

Los Gatos, CA • lupified@gmail.com

SUMMARY

Applied Mathematics PhD researcher developing **physically consistent AI emulators and pipelines for complex, multimodal, time-dependent systems**. Specializing in autoregressive deep learning for weather, ocean, and turbulence modeling: long-rollout stability, spectral fidelity, generative data assimilation, and real-time multimodal event detection. First-author publications spanning AI emulation, plasma control, and detector physics; experienced driving independent research from concept to publication.

SELECTED RESEARCH & PROJECTS

Physically Consistent Regional Ocean Emulation | Fujitsu Research of America 2024–2025

- Developed a deep learning framework for autoregressive emulation of Gulf of Mexico sea-surface dynamics; novel **spectral loss objective** enables stable, physically consistent rollouts at 8 km resolution over decadal timescales without unphysical drift.
- Coupled emulator with a physics-constrained generative model to simultaneously downscale and bias-correct predictions to 4 km resolution. Published (first author) in JGR: Machine Learning and Computation. doi:10.1029/2025JH000851

Multimodal Kilonova Detection Pipeline | Los Alamos National Laboratory 2026

- Built a pipeline for live identification of kilonova astronomical events in survey telescopes, fusing multimodal data streams for real-time detection of rare transients.

Generative Data Assimilation Under Extreme Sparsity 2025

- Reconstructed full ocean and turbulence states from sparse satellite and Lagrangian observations by conditioning **denoising diffusion models on neural operator outputs**; accurate small-scale, high-wavenumber recovery at up to 99.9% observational sparsity. Published in Machine Learning: Earth. doi:10.1088/3049-4753/ae0b70

Full-Field Reconstruction & Equation Discovery in Nonlinear Systems Ongoing

- Developed framework recovering full spatiotemporal fields from a time-series signal at a single spatial point in cryogenic soliton propagation; recovered fields used to identify higher-order terms in the underlying system PDE.

Spectral Theory of Diffusion-Based Reconstruction for Turbulent Flows Ongoing

- Investigating how noise schedules and turbulence energy profiles govern the scale-dependent reconstruction accuracy of diffusion generative models across large- and small-scale features.

EXPERIENCE

Research Intern, Los Alamos National Laboratory 2026

Designed a multimodal machine learning pipeline for live kilonova event identification across survey telescope data streams (see Projects).

AI Geophysical Researcher, Fujitsu Research of America 2024

Developed AI frameworks for regional sea-surface emulation in the Converging Technologies Lab: autoregressive neural operator forecasting, physics-constrained generative downscaling, and stability strategies for long-horizon rollout; first-author journal publication. Contributed physics-informed neural operator wave forecasting and sparse data assimilation to SWR-Viz, an AI-assisted ship weather-routing and emissions analytics pipeline (published, ISVC 2025).

Consultant Data Scientist, Ataraxia Therapeutics | part-time 2023–2025

Analyzed multichannel EEG time series and identified synaptic spike patterns to support neurological disease treatment development.

Bioinformatics Programmer, UCSF Pathology, Data Science CoLabs & ImmunoX 2018–2021

Characterized the immune cell environment across heterogeneous cancer types to support novel drug and therapy development, integrating multimodal biological data: RNA sequencing, single-cell transcriptomics, flow cytometry, and microscopy statistics.

Scientific Engineering Associate, Lawrence Berkeley National Laboratory 2018

Characterized radiation signatures in aerial measurement data collected near the Fukushima nuclear reactor meltdown.

Machine Learning Scientist, Princeton Plasma Physics Laboratory 2016–2018

Researched machine learning methods for real-time plasma disruption prediction in magnetic confinement fusion under Prof. Egemen Kolemen. APS poster presentation; publication in Physics of Plasmas.

Consultant Statistical Analyst, Cognito Therapeutics 2017–2018

Mined and analyzed Medicare datasets and consumer-trial data for Cognito's Alzheimer's disease treatment product.

Applied Physics Research Assistant, nEXO, Stanford University

2013–2016

Researched novel silicon photon detectors under Prof. Giorgio Gratta: independent experimental design and upgrades, simulation, and data analysis in Python; publication in IEEE Transactions on Nuclear Science.

EDUCATION

Ph.D. Applied Mathematics, University of California, Santa Cruz

2023–present

Theoretical and Applied Complex Systems Lab; advisor: Prof. Ashesh Chattopadhyay. Dissertation focus: deep learning emulation of nonlinear geophysical systems: autoregressive stability, spectral accuracy, and generative data assimilation.

M.S. Physics, San Francisco State University

2021–2023

Research in geophysical fluid dynamics and machine learning.

B.S. Physics, Stanford University

2013–2017

Minors in Mathematics and Psychology.

PUBLICATIONS

- Hazarika, S., **Lupin-Jimenez, L.**, Vuppala, R., Chattopadhyay, A., & Wong, H. Y. “SWR-Viz: AI-Assisted Interactive Visual Analytics Framework for Ship Weather Routing.” *Advances in Visual Computing (ISVC 2025), Lecture Notes in Computer Science*, vol. 16397, Springer (2026).
- **Lupin-Jimenez, L.**, Darman, M., Hazarika, S., Wu, T., Gray, M., He, R., Wong, A., & Chattopadhyay, A. “Simultaneous Emulation and Downscaling With Physically Consistent Deep Learning-Based Regional Ocean Emulators.” *Journal of Geophysical Research: Machine Learning and Computation*, 2(3), e2025JH000851 (2025).
- Asefi, N., **Lupin-Jimenez, L.**, Wu, T., He, R., & Chattopadhyay, A. “Generative Lagrangian Data Assimilation for Ocean Dynamics Under Extreme Sparsity.” *Machine Learning: Earth*, 1, 015009 (2025).
- “Activating Immune Recognition in Pancreatic Ductal Adenocarcinoma Using Autophagy Inhibition, MEK Blockade and CD40 Agonism.” *bioRxiv* (2020).
- “Machine Learning Control for Disruption and Tearing Mode Avoidance.” *Physics of Plasmas*, 27(2) (2020).
- “VUV-Sensitive Silicon Photomultipliers for Xenon Scintillation Light Detection in nEXO.” *IEEE Transactions on Nuclear Science*, 65(11), 2823–2833 (2018).

TECHNICAL SKILLS

Machine Learning: PyTorch, TensorFlow • diffusion generative models, neural operators, autoregressive & sequence modeling • physics-informed losses, spectral methods • multimodal data fusion • data assimilation

Scientific Computing: Python, C++, Fortran, Mathematica • statistical analysis • data visualization

Other: JavaScript, CAD design, circuitry, physical experimental design

LEADERSHIP & MENTORING

Graduate Teaching Assistant, UC Santa Cruz: Introductory Laboratory Mechanics, Multivariate Calculus, and Partial Differential Equations.

AI instructor for middle/high school classrooms, Inspirit AI (2021–2024, online and international in-person) • private STEM tutor, Grade Potential (2017–2020) • volunteer STEM teaching and community events (Science in Service, Cardinal Scholars) • music and arts festival organizer.

LANGUAGES & INTERESTS

Languages: English (native), Spanish (conversant), Japanese (basic).

Interests: science fiction, mountain biking, lacrosse, hiking, camping, dance, robotics, violin, sitar.

References available upon request.